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Appendix A

User Guide

A.1 Installing TexLive

A.1.1 Linux

- for full TexLive installation on linux, use below;

```
sudo apt install texlive-scheme-full
```

- The installation may take few minutes to complete.
- Alternatively, you can download 'install-tl' file from ctan systems
- Type perl . Ensure you have perl installed
- Then perl install-tl -gui
- finally, perl install-tl -help

A.1.2 MacOs

- Download MaxTex package from tug.org
- Once downloaded, double-click on the downloaded file and follow the installation instructions. For more information on installation, visit tug.org

A.1.3 Windows

- Download install-tl-windows.exe file from ctan systems
- Open the file and follow the installation instructions

A.2 Installing Bash for Windows

- Visit msys2 or git for more information on installing bash for windows.

A.3 Structure of Main TEX File

- User should ensure that their main TEX document matches this structure for seamless integration. The document class used is **article**.

Listing A.1: Structure of the main TEX file.

```
\usepackage{standalone} % indicate standalone
\usepackage{graphicx}
\usepackage{tikz}
% include all the libraries in subfile
\usetikzlibrary{graphs}
\usetikzlibrary{shapes.geometric}
\begin{document}
\begin{figure}[!h]
    \centering
    \resizebox{0.5\textwidth}{!}{\input{tikz.tex}}% add TEX file containing tikz code
\end{figure}%
\end{document}
```

A.4 Writing JSON file

This section discusses each key of the JSON specification and its possible values.

```
{
     "orientation": "dt",
     "shapeFill": "red",
     "fontColor": "white",
     "shapeColor": "blue",
    "idx1": {
        "label": "Start",
        "type": "terminator",
```

```
        "next": [
            {
                "id": "idx2"
            }
        ],
        "shapeFill": "red",
        "fontColor": "white",
        "shapeColor": "blue",
    },
    "idx2": {
        "label": "Input",
        "type": "io",
        "next": [
            {
                "id": "idx3"
            }
        ],
        "shapeFill": "blue"
    }
}
```

Global Properties

- **orientation**: determines the direction of the graph. It takes values, td (top-down/bottom), lr (left-right), dt (down/bottom-top) and rl (right-left).
- **shapeFill**: changes the background contour colour of all vertices to the specified colour. It accepts all colours available in latex.
- **shapeColor**: changes the contour colour of all vertices to the specified colour.
- **fontColor**: changes the font colour of all vertices to the specified colour.

Creating a Vertex

- **idx**: the unique id provided by the user. It does not always have to be "idx". It can be any text but must be unique for different vertices.

- **label**: the label of the vertex.
- **type**: the flowchart vertex type. Valid values include, terminator, process, io and decision.
- **next**: array of vertices this vertex is connected to. The ids of the vertices connected are used before they are created. If a referenced vertex is not created when indicated as the next vertex, a default vertex will be automatically created. A default vertex is represented as a terminator with the label "default".
- **shapeFill**: changes the background contour colour of a particular vertex to the specified colour.
- **shapeColor**: changes the contour colour of a particular vertex to the specified colour.
- **fontColor**: changes the font colour of a particular vertex to the specified colour.